

NumPy Complete Placement Cheat Sheet

Import

`import numpy as np`

Array Creation

`np.array(), zeros(), ones(), empty(), full(), eye(), identity(), arange(), linspace()`

Properties

`shape, ndim, size, dtype, astype()`

Indexing

`arr[0], arr[-1], arr[1:4], arr[:,1], arr[1:], arr[::-1]`

Advanced Indexing

Fancy: `arr[[0,2]]`

Boolean: `arr[arr>50]`

`np.where(cond,x,y)`

Views

`view()` shares memory

`copy()` creates new memory

`flatten()` = copy

`ravel()` = view if possible

Broadcasting

Rules:

1. Compare shapes right→left
2. Equal or one dimension is 1
3. Else `ValueError`

Math

`add, subtract, multiply, divide, power, mod, sqrt, square, abs, exp, log, sin, cos, tan`

Aggregation

`sum, mean, median, min, max, argmin, argmax, std, var`
`axis=0` columns, `axis=1` rows

Reshape

reshape(), reshape(-1,1), reshape(1,-1), transpose(.T), resize(), expand_dims(), squeeze()

Join/Split

concatenate(), stack(), vstack(), hstack(), split(), vsplit(), hsplit()

Random

rand(), randn(), randint(), choice(), shuffle(), seed()

Interview

shape vs size vs ndim
view vs copy
flatten vs ravel
stack vs concatenate
argmax vs max
broadcasting rules

Memory Tricks

VSTACK=Vertical
HSTACK=Horizontal
arg=Index
flatten=Copy
ravel=View
Broadcast=Stretch not Copy